

THE ZINC-FINGER STRUCTURE OF THE CYTOSOLIC DOMAIN OF THE CCHFV Gn PROTEIN IS CONSERVED IN ITS MEMBRANE-BOUND FORM

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Here we characterize the structure of the cytosolic domain of the Crimean-Congo hemorrhagic fever virus (CCHFV) Gn protein in its membrane-bound form.

CCHFV is emerging in Europe, as its main vector, the Hyalomma tick, is spreading northward due to global warming. With a case fatality rate of up to 40%, CCHFV is one of the deadliest viruses. Despite this, there is no approved vaccine or treatment. The Gn protein is crucial for viral RNA packaging upon assembly. The structure of its isolated cytosolic domain is known; it features a zinc finger, but the membrane-bound form remains uncharacterized.

We aim to elucidate the structure of the membrane-bound state using a synergistic combination of cell-free protein synthesis (CFPS) and solid-state NMR, ideal for membrane-associated domains not accessible by cryoEM. To overcome challenges of membrane protein expression, « divide-and-conquer » strategies are commonly used to study soluble domains. However, the membrane environment can significantly influence protein structure and dynamics. CFPS is highly effective at producing membrane proteins in their native form and enables selective isotope labeling, which provides spectral resolution.

Our results show that the characteristic zinc-finger motif, previously identified in isolation, is conserved in the membrane-bound form. They also suggest the existence of two conformational states, indicating either a partially asymmetric multimer, or a second distinct conformation. Beyond its biological significance, Gn serves as a model to develop powerful biochemical and NMR tools, paving the way for the structural characterization of equally complex systems, such as virulence factors and vaccine targets.

REFERENCES

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